Sri Lanka Institute of Information Technology

IT3021 - Data warehousing and Business Intelligence

Assignment 02

IT23815728

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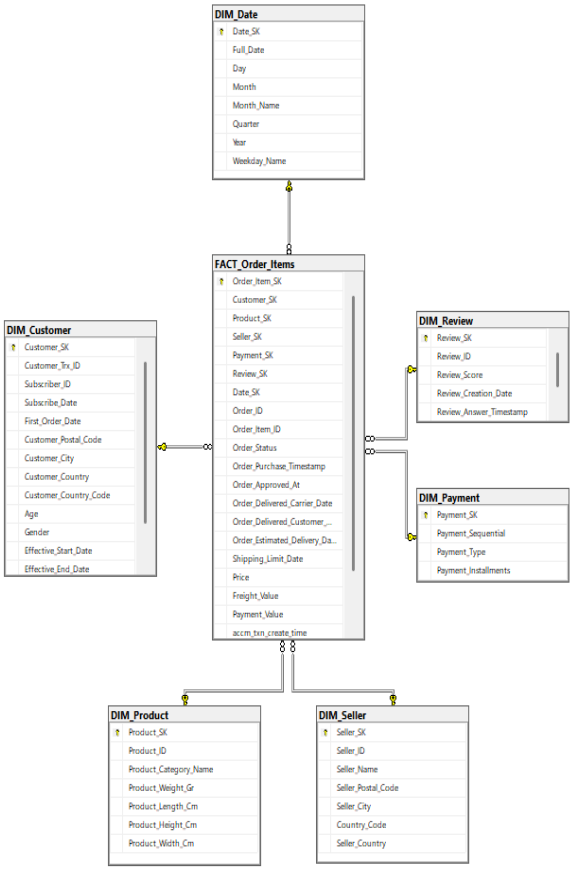
Data source for the assignment 2



- **DIM_Date** stores all date-related attributes used for time-based analysis, including Date_SK as the surrogate key, along with Full_Date, Day, Month, Month_Name, Quarter, Year, and Weekday_Name. This dimension enables the drill-down hierarchy from Year → Quarter → Month used in your Power BI report, and supports time intelligence filtering across all fact table transactions.

- **DIM_Customer** captures the full profile of each customer, with Customer_SK as the surrogate key and business key Customer_Tr_ID. It includes demographic and geographic attributes such as Gender, Age, Customer_City, Customer_Country, and Customer_Postal_Code, as well as SCD (Slowly Changing Dimension) tracking fields

Effective_Start_Date and Effective_End_Date, indicating this dimension is designed as a Type 2 SCD to preserve historical customer changes.

- **DIM_Product** describes the items available for purchase, identified by Product_SK and Product_ID. It contains classification details like Product_Category_Name and physical specifications including Product_Weight_G, Product_Length_Cm, Product_Height_Cm, and Product_Width_Cm, making it useful for both sales analysis and logistics or shipping cost calculations.

- **DIM_Seller** holds information about the merchants or sellers operating on the platform, with Seller_SK as the surrogate key and Seller_ID as the natural key. Geographic attributes such as Seller_Name, Seller_Postal_Code, Seller_City, Country_Code, and Seller_Country allow for regional performance analysis and seller-level reporting.

- **DIM_Payment** describes the payment methods associated with orders, using Payment_SK as the surrogate key. It includes Payment_Sequential, Payment_Type, and Payment_Installments, which together allow analysis of how customers choose to pay — whether in full or across multiple installments — and supports segmentation by payment behavior.

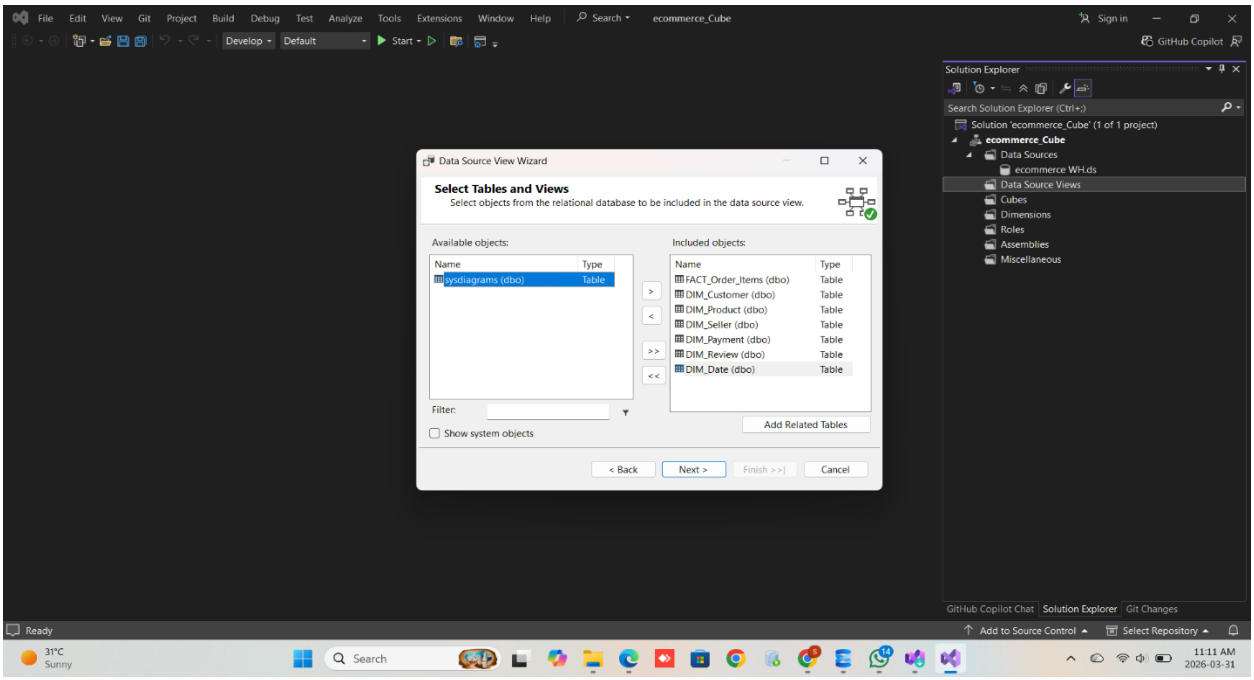
- **DIM_Review** stores customer feedback linked to orders, identified by Review_ID and joined to the fact table via Review_SK. It captures Review_Score, Review_Creation_Date, and Review_Answer_Timestamp, enabling customer satisfaction analysis, response time tracking, and correlation of review scores with order outcomes.

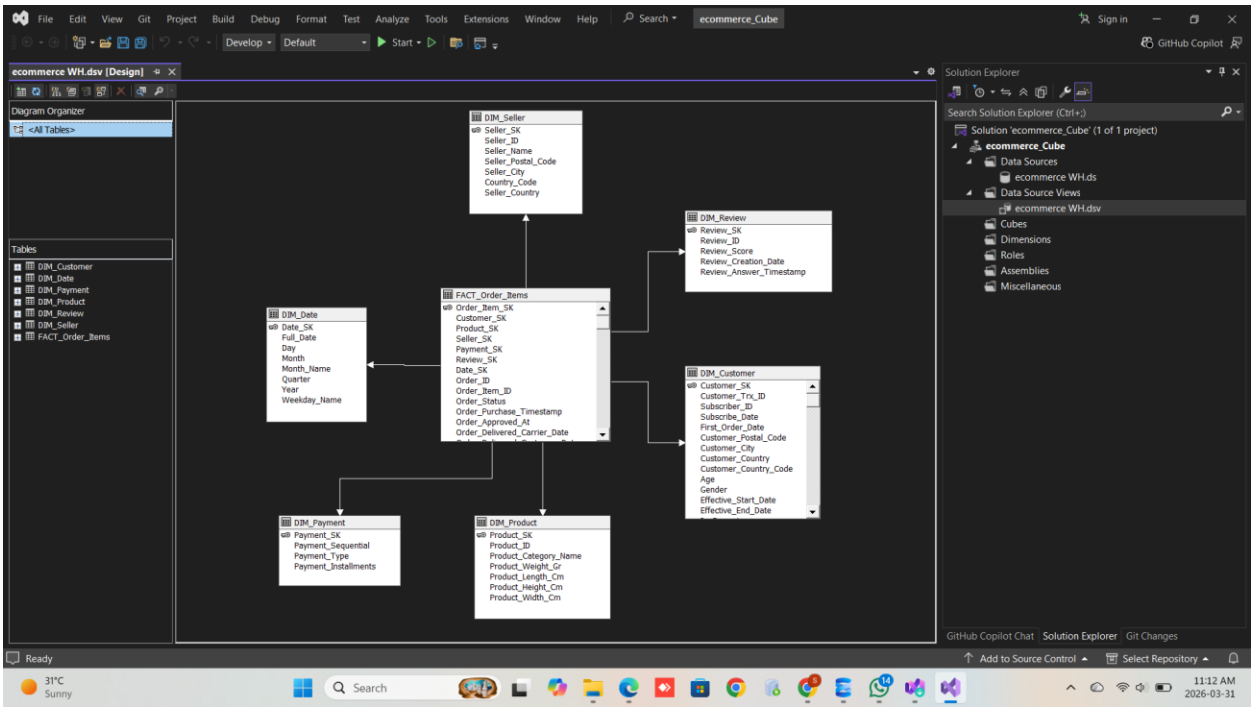
- **FACT_Order_Items** is the central fact table in the schema, recording every individual order line item transacted on the platform. It contains foreign keys linking to all six dimension tables — Customer_SK, Product_SK, Seller_SK, Payment_SK, Review_SK, and Date_SK — and stores measurable facts such as Price, Freight_Value, and Payment_Value. Operational attributes like Order_Status, Order_Purchase_Timestamp, Order_Approved_At, Order_Delivered_Carrier_Date, Order_Delivered_Customer_Date, Order_Estimated_Delivery_Date, and Shipping_Limit_Date make it the primary source for both financial reporting and supply chain performance analysis.

SSAS Cube implementation

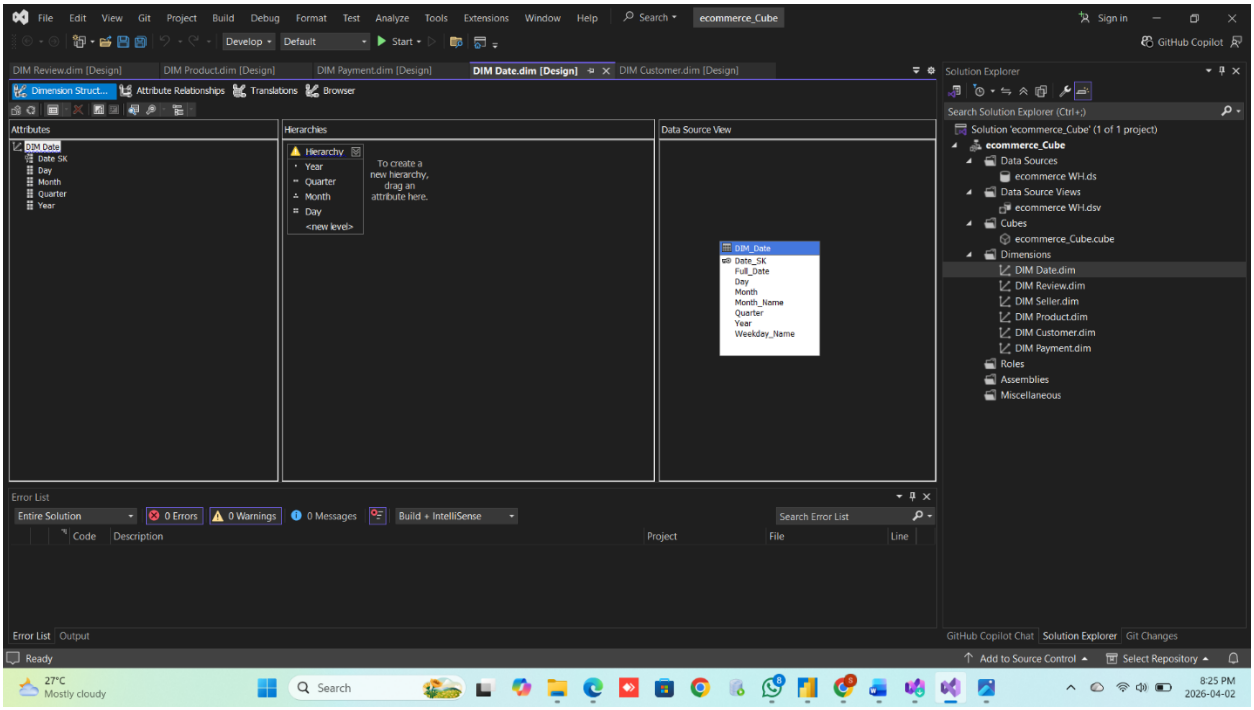
- Opened Visual Studio and created a new "Analysis Services Multidimensional and Data Mining Project" to start building the SSAS cube.

- A new Data Source View was created using the data warehouse connection. All required dimension tables and the fact table were added as included objects.



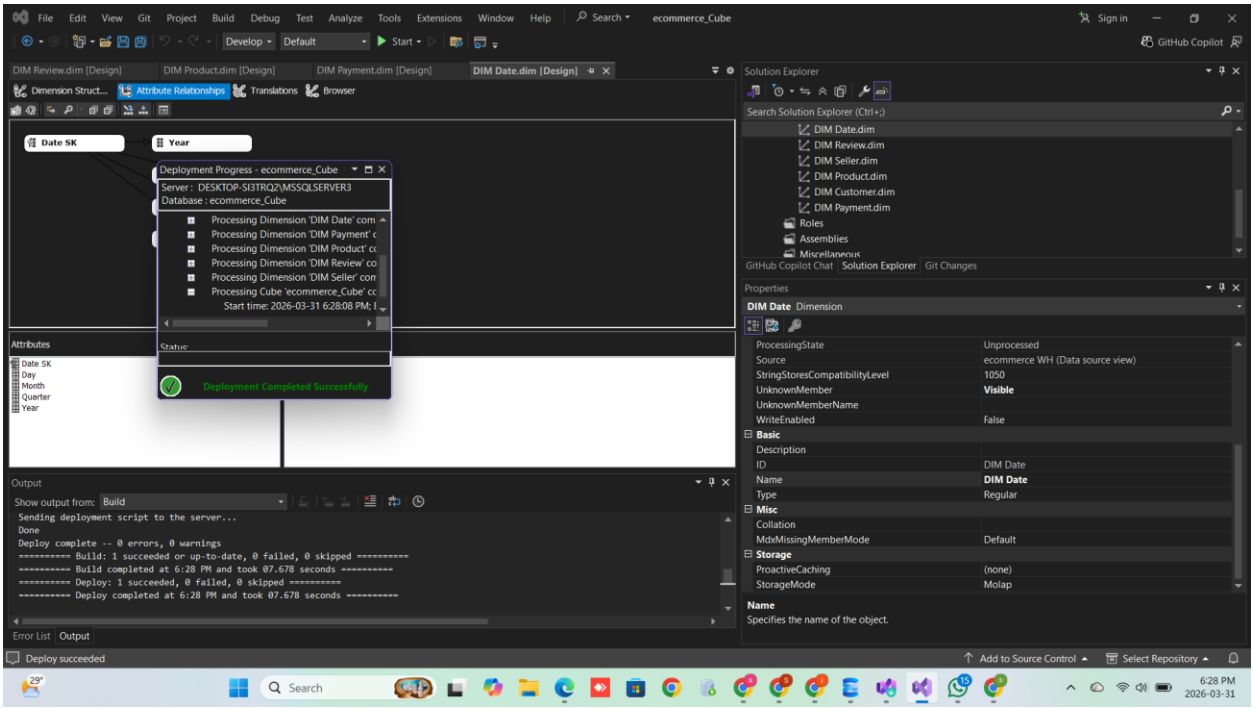


- A hierarchy was created in the Date dimension by arranging the Year, Quarter, Month, and Day attributes in logical order to support drill-down analysis.



- The Date hierarchy was manually created by mapping the Year, Quarter, Month, and Day attributes in the correct order within the Date dimension structure.

- The cube was deployed to the SQL Server Analysis Services instance to create the necessary metadata and structure on the server.

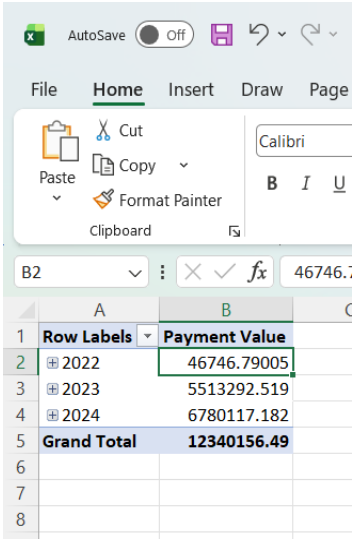


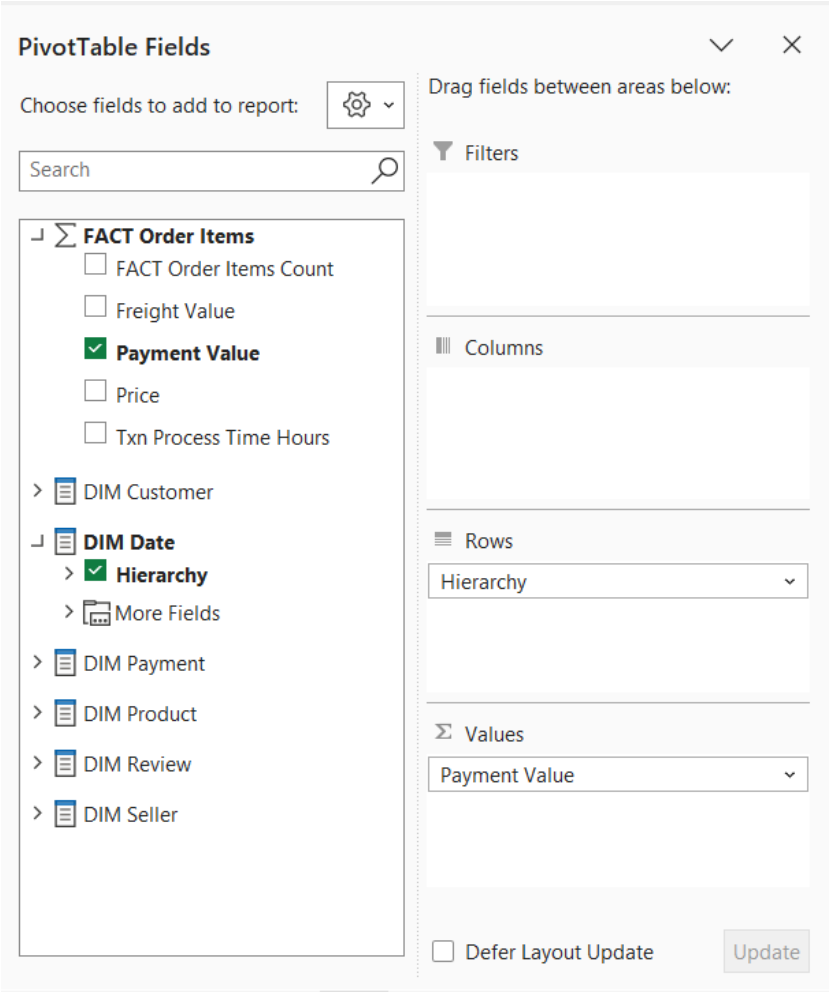
Demonstration of OLAP operations

Connected Excel to SSAS Cube Used Excel's Get Data feature to establish a connection to the deployed ecommerce_Cube on the SSAS server DESKTOP-SI3TRQ \MSSQLSERVER3. Selected the ecommerce_Cube database and inserted a PivotTable report linked to the cube.

Roll up

Roll-up Aggregated sales data from lower levels to higher levels using the Date Hierarchy. Daily transaction records were rolled up to Monthly totals, and Monthly totals were further aggregated to Yearly summaries. For example, individual order amounts across all days were summarized to show Total Revenue per Year, demonstrating how detailed transactional data can be consolidated into high-level business summaries.





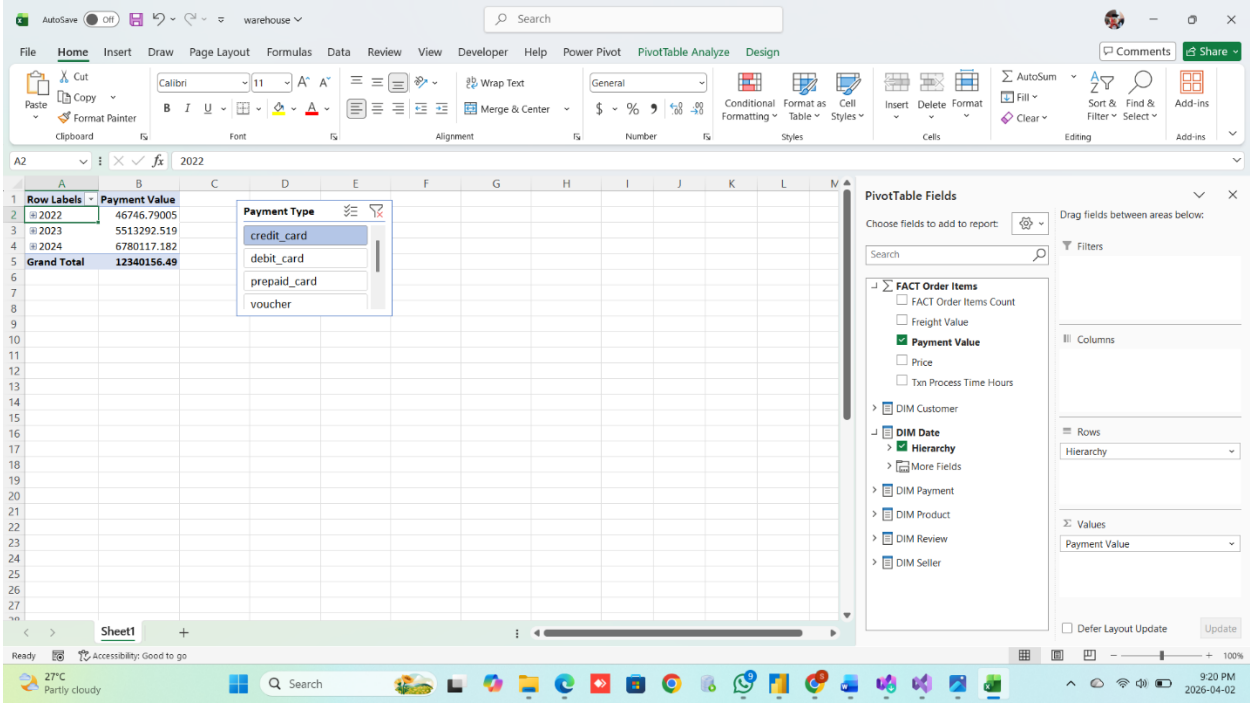
Drill-Down

Drill-down Navigated from high-level yearly summaries down to granular monthly details using the Date Hierarchy (Year → Quarter → Month → Day). By clicking the expanding (+) button next to a Year value in the PivotTable, the data was broken down into Quarters and further expanded to show individual Month-level Total Revenue figures. This allowed detailed examination of seasonal sales patterns within each year.

Row Labels	Payment Value
⊟ 2022	
⊞ 3	211.2899933
⊞ 4	46535.50006
⊟ 2023	
⊞ 1	663158.1992
⊞ 2	1133217.49
⊞ 3	1526395.69
⊞ 4	2190521.14
⊟ 2024	
⊞ 1	2551417.309
⊞ 2	2646097.972
⊞ 3	1582601.9
Grand Total	**12340156.49**

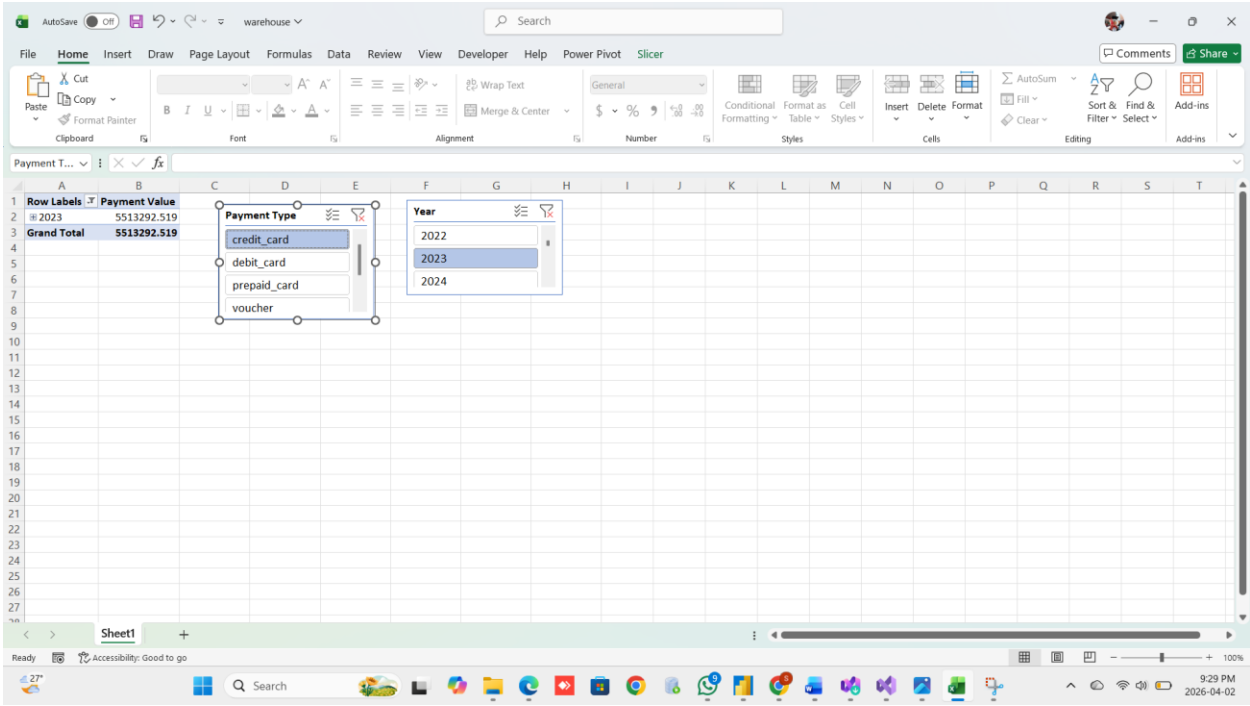
Slice Slice

Filtered the PivotTable on a single dimension value using an Excel Slicer. A slicer was inserted on the Payment Type from Dim Payment, and a single Payment Type (e.g. "Credit_card") was selected. The PivotTable automatically filtered to display Total Payment for Credit_card payments only across all time periods, isolating a single dimension for focused analysis.



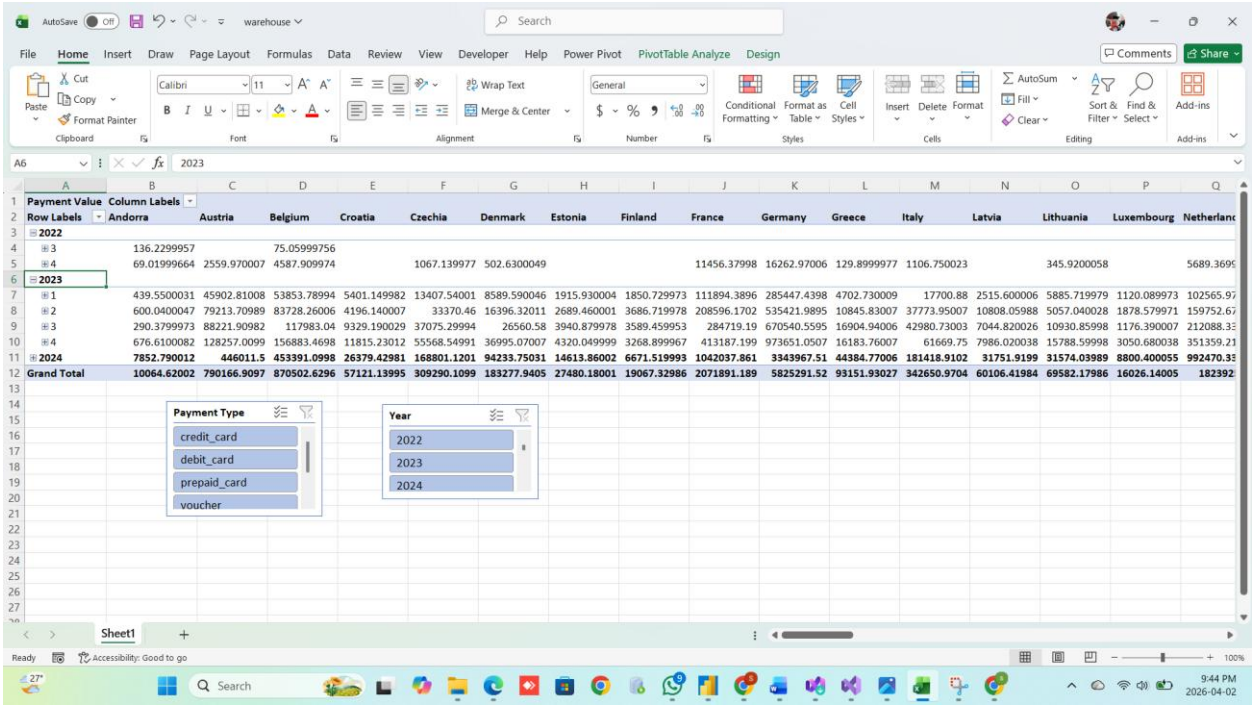
Dice

Dice Applied simultaneous filters across two dimensions to analyze a specific data subcube. Two slicers were used together — one filtering by Payment Type (e.g. " Credit_card ") and another filtering by Year (e.g. "2023"). The PivotTable displayed Total Payment only for credit card payments in 2023, demonstrating how multiple dimension filters can be combined to extract a precise subset of the data.



Pivot Table

The data axes of the PivotTable were rotated to reorganize the analytical perspective. The Country dimension was moved from the Columns area to the Rows area while Year remained in Rows, resulting in a cross-tabular view where each row represents a Year and each column represents a Customer Country. Two slicers — Payment Type (credit card, debit card, prepaid card, voucher) and Year (2022, 2023, 2024) — remain connected to the table, enabling dynamic filtering without altering the underlying data. This rearrangement provides a different analytical viewpoint, making it easier to compare payment values across multiple European markets such as Germany, France, Italy, and Latvia simultaneously while maintaining year-on-year visibility.

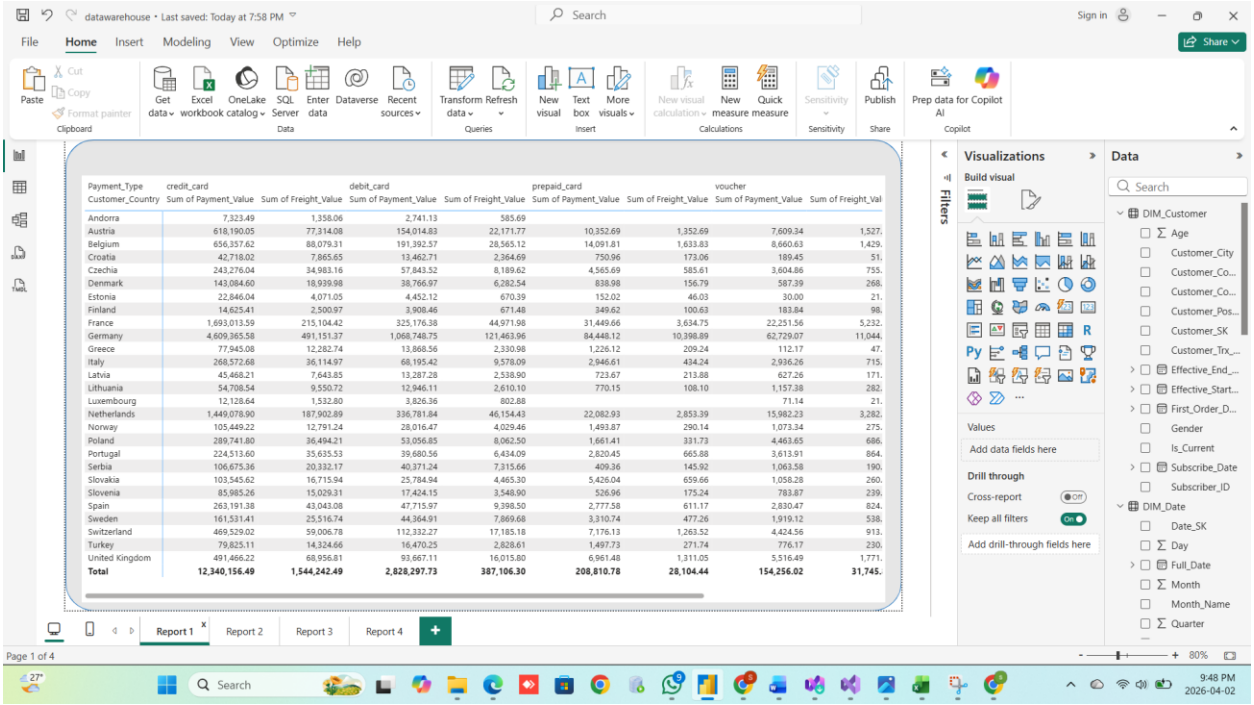


Payment Value	Column Labels															
Row Labels	Andorra	Austria	Belgium	Croatia	Czechia	Denmark	Estonia	Finland	France	Germany	Greece	Italy	Latvia	Lithuania	Luxembourg	Netherlanc
2022																
3	136.2299957		75.05999756													
4	69.01999664	2559.970007	4587.909974		1067.139977	502.6300049			11456.37998	16262.97006	129.8999977	1106.750023		345.9200058		5689.369
2023																
1	439.5500031	45902.81008	53853.78994	5401.149982	13407.54001	8589.590046	1915.930004	1850.729973	111894.3896	285447.4398	4702.730009	17700.88	2515.600006	5885.719979	1120.089973	102565.97
2	600.0400047	79213.70989	83728.26006	4196.140007	33370.46	16396.32011	2689.460001	3686.719978	208596.1702	535421.9895	10845.83007	37773.95007	10808.05988	5057.040028	1878.579971	159752.67
3	290.3799973	88221.90982	117983.04	9329.190029	37075.29994	26560.58	3940.879978	3589.459953	284719.19	670540.5595	16904.94006	42980.73003	7044.820026	10930.85998	1176.390007	212088.33
4	676.6100082	128257.0099	156883.4698	11815.23012	55568.54991	36995.07007	4320.049999	3268.899967	413187.199	973651.0507	16183.76007	61669.75	7986.020038	15788.59998	3050.680038	351359.21
2024	7852.790012	446011.5	453391.0998	26379.42981	168801.1201	94233.75031	14613.86002	6671.519993	1042037.861	3343967.51	44384.77006	181418.9102	31751.9199	31574.03989	8800.400055	992470.33
Grand Total	10064.62002	790166.9097	870502.6296	57121.13995	309290.1099	183277.9405	27480.18001	19067.32986	2071891.189	5825291.52	93151.93027	342650.9704	60106.41984	69582.17986	16026.14005	182392

PowerBI Reports

Report 1

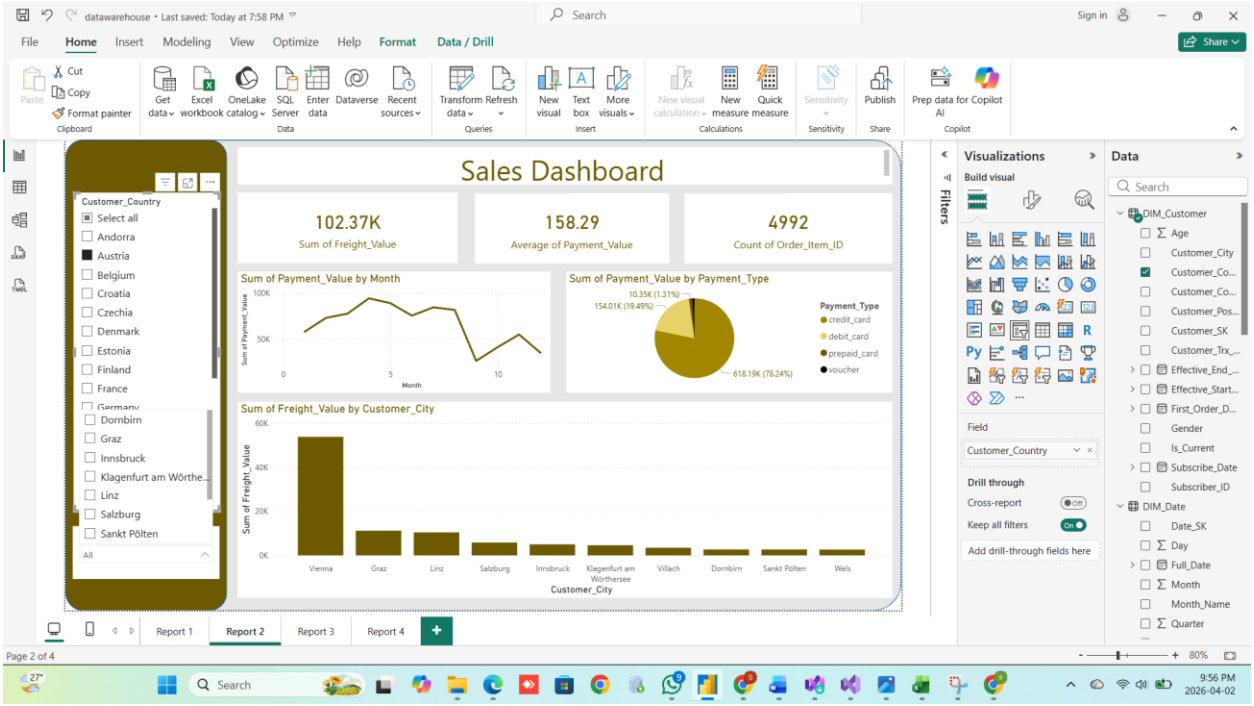
A Matrix visual was created in Power BI to display both Sum of Payment Value and Sum of Freight Value across each Customer Country, broken down by Payment Type (credit card, debit card, prepaid card, and voucher). Each column pair represents a payment method, allowing side-by-side comparison of payment and freight costs per country. High-value markets such as Germany, France, Netherlands, and the United Kingdom are clearly visible as dominant regions. The Total row at the bottom consolidates all countries, providing an overall summary that makes it easy to identify dominant payment behaviors and freight cost distributions across European regions.



Payment_Type	credit_card		debit_card		prepaid_card		voucher	
Customer_Country	Sum of Payment_Value	Sum of Freight_Value	Sum of Payment_Value	Sum of Freight_Value	Sum of Payment_Value	Sum of Freight_Value	Sum of Payment_Value	Sum of Freight_Val
Andorra	7,323.49	1,358.06	2,741.13	585.69				
Austria	618,190.05	77,314.08	154,014.83	22,171.77	10,352.69	1,352.69	7,609.34	1,527.
Belgium	656,357.62	88,079.31	191,392.57	28,565.12	14,091.81	1,633.83	8,660.63	1,429.
Croatia	42,718.02	7,865.65	13,462.71	2,364.69	750.96	173.06	189.45	51.
Czechia	243,276.04	34,983.16	57,843.52	8,189.62	4,565.69	585.61	3,604.86	755.
Denmark	143,084.60	18,939.98	38,766.97	6,282.54	838.98	156.79	587.39	268.
Estonia	22,846.04	4,071.05	4,452.12	670.39	152.02	46.03	30.00	21.
Finland	14,625.41	2,500.97	3,908.46	671.48	349.62	100.63	183.84	98.
France	1,693,013.59	215,104.42	325,176.38	44,971.98	31,449.66	3,634.75	22,251.56	5,232.
Germany	4,609,365.58	491,151.37	1,068,748.75	121,463.96	84,448.12	10,398.89	62,729.07	11,044.
Greece	77,945.08	12,282.74	13,868.56	2,330.98	1,226.12	209.24	112.17	47.
Italy	268,572.68	36,114.97	68,195.42	9,578.09	2,946.61	434.24	2,936.26	715.
Latvia	45,468.21	7,643.85	13,287.28	2,538.90	723.67	213.88	627.26	171.
Lithuania	54,708.54	9,550.72	12,946.11	2,610.10	770.15	108.10	1,157.38	282.
Luxembourg	12,128.64	1,532.80	3,826.36	802.88			71.14	21.
Netherlands	1,449,078.90	187,902.89	336,781.84	46,154.43	22,082.93	2,853.39	15,982.23	3,282.
Norway	105,449.22	12,791.24	28,016.47	4,029.46	1,493.87	290.14	1,073.34	275.
Poland	289,741.80	36,494.21	53,056.85	8,062.50	1,661.41	331.73	4,463.65	686.
Portugal	224,513.60	35,635.53	39,680.56	6,434.09	2,820.45	665.88	3,613.91	864.
Serbia	106,675.36	20,332.17	40,371.24	7,315.66	409.36	145.92	1,063.58	190.
Slovakia	103,545.62	16,715.94	25,784.94	4,465.30	5,426.04	659.66	1,058.28	260.
Slovenia	85,985.26	15,029.31	17,424.15	3,548.90	526.96	175.24	783.87	239.
Spain	263,191.38	43,043.08	47,715.97	9,398.50	2,777.58	611.17	2,830.47	824.
Sweden	161,531.41	25,516.74	44,364.91	7,869.68	3,310.74	477.26	1,919.12	538.
Switzerland	469,529.02	59,006.78	112,332.27	17,185.18	7,176.13	1,263.52	4,424.56	913.
Turkey	79,825.11	14,324.66	16,470.25	2,828.61	1,497.73	271.74	776.17	230.
United Kingdom	491,466.22	68,956.81	93,667.11	16,015.80	6,961.48	1,311.05	5,516.49	1,771.
Total	**12,340,156.49**	**1,544,242.49**	**2,828,297.73**	**387,106.30**	**208,810.78**	**28,104.44**	**154,256.02**	**31,745.**

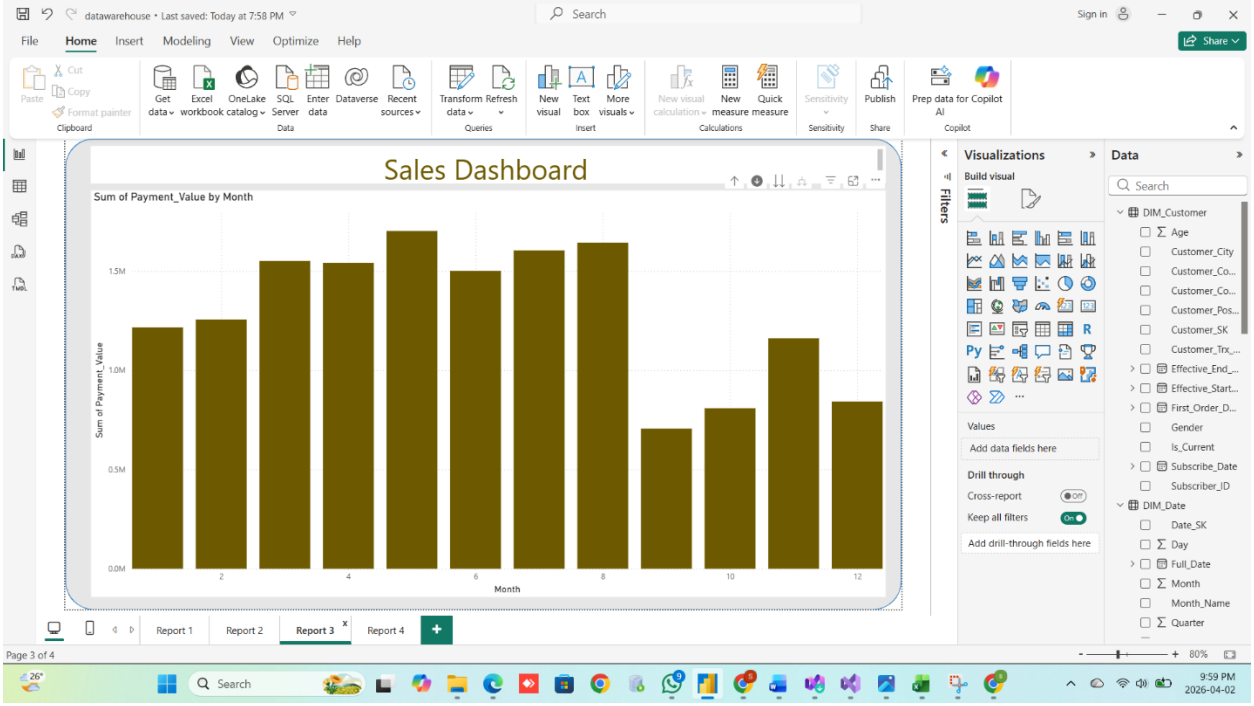
Report 2

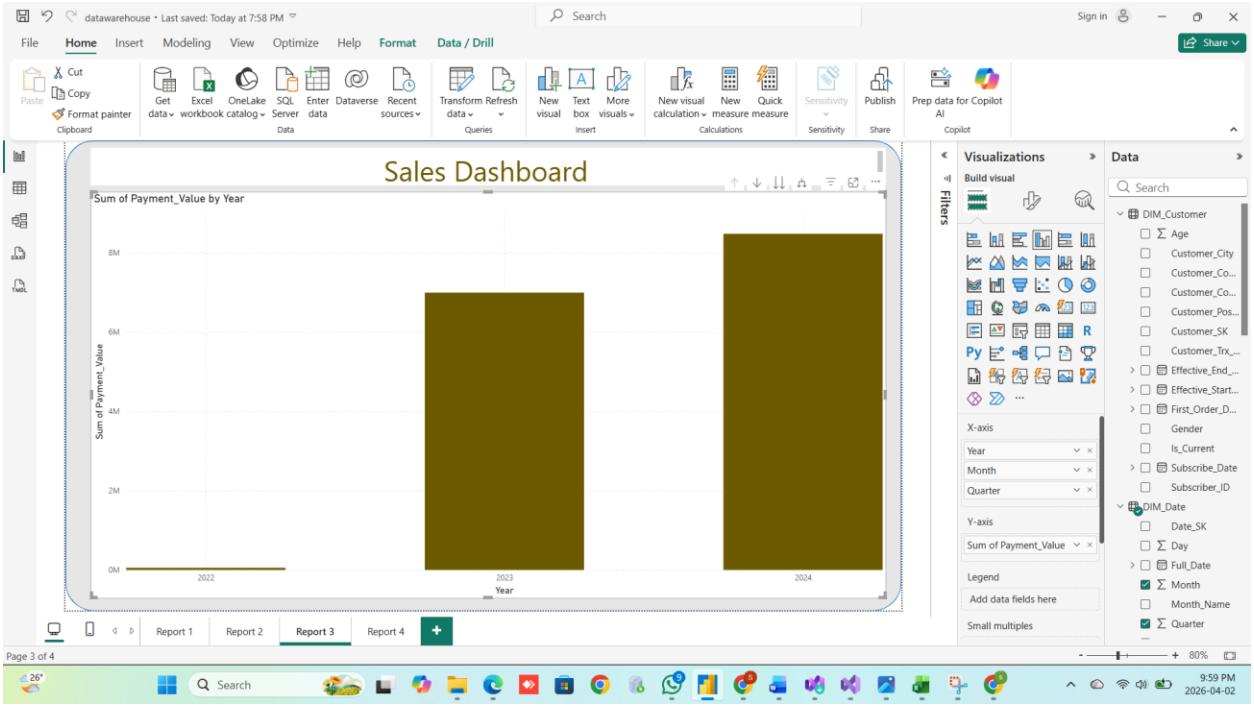
Report 2 demonstrates cascading filters using two interactive slicers. The first slicer filters by Customer Country and the second slicer dynamically updates to show only the Customer Cities belonging to the selected country. This cascading filtering behavior ensures that users can narrow down their selection in a logical top-down manner. The report also includes a Line Chart showing Sum of Payment Value by Month and a Pie Chart showing Sum of Payment Value by Payment Type. Additionally, a Bar Chart displays the Sum of Freight Value by Customer City, providing a geographic breakdown of freight distribution. When a country is selected in the first slicer, all visuals on the page including both charts, the KPI cards, and the city slicer are automatically filtered to reflect only the relevant data for that selection.

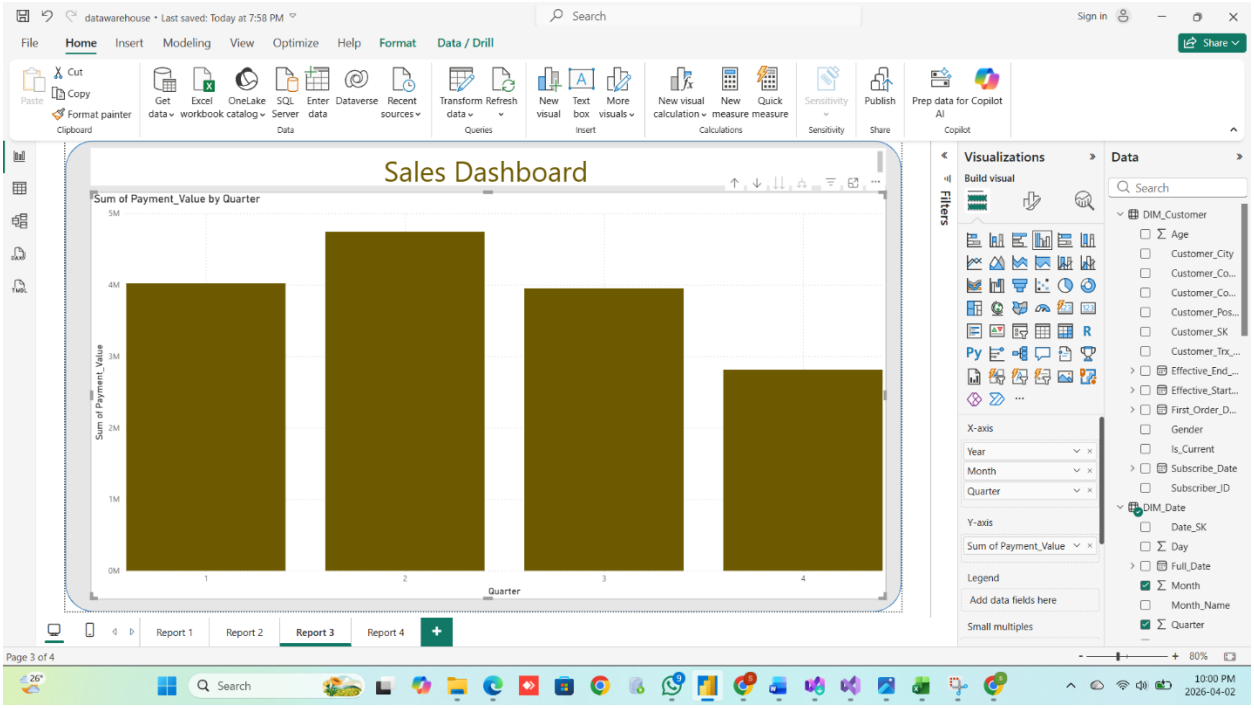


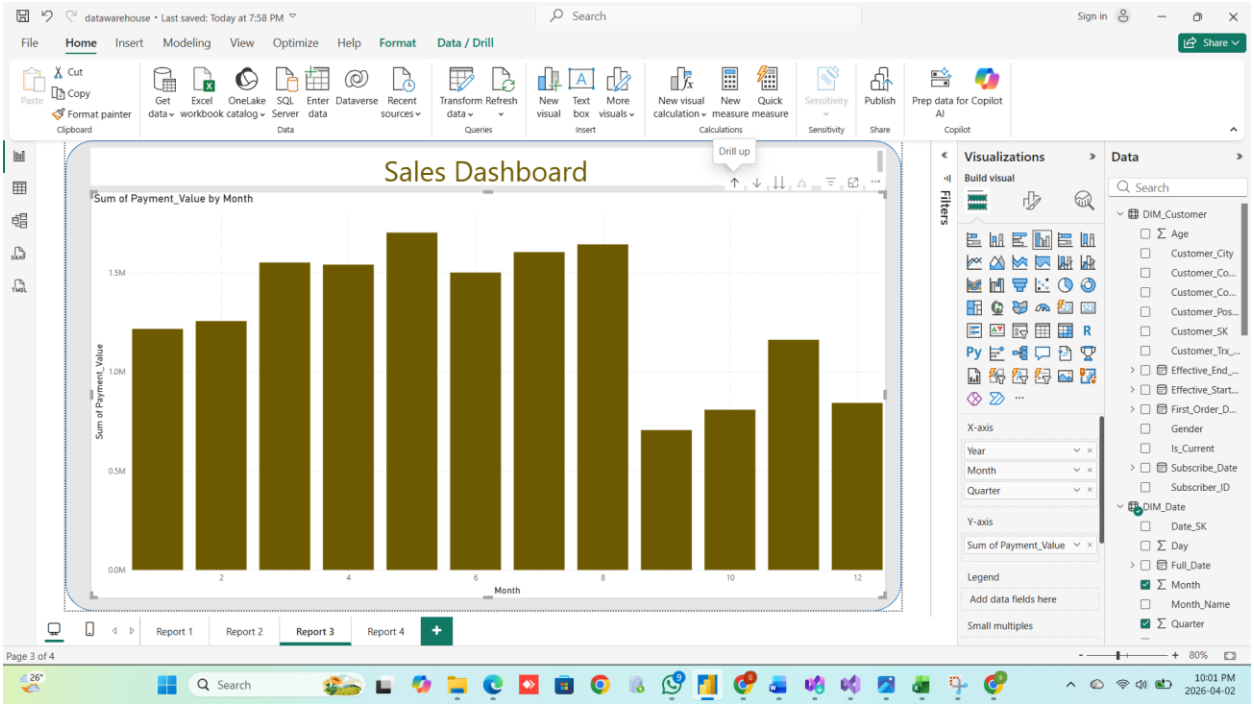
Report 3

Report 3 demonstrates hierarchical drill-down functionality using a Clustered Column Chart. The X-axis is configured with a date hierarchy consisting of months across the year, and the Y-axis displays the Sum of Payment Value. Users can drill down through the hierarchy by clicking on individual bars in the chart, allowing navigation from a yearly summary level down to quarterly and then monthly payment details. The drill-up button allows users to navigate back up the hierarchy seamlessly. The chart reveals noticeable fluctuations in payment values across months, with mid-year months showing relatively higher performance compared to the latter part of the year. This report enables time-based trend analysis at multiple levels of granularity, supporting business decisions based on seasonal and monthly payment patterns.




Sales Dashboard
Sum of Payment_Value by Year
Sum of Payment_Value
8M
6M
4M
2M
0M
2022
2023
2024
Year


Sales Dashboard
Sum of Payment_Value by Quarter
Sum of Payment_Value
5M
4M
3M
2M
1M
0M
1
2
3
4
Quarter


Sales Dashboard
Sum of Payment_Value by Month
Sum of Payment_Value
1.5M
1.0M
0.5M
0.0M
2
4
6
8
10
12
Month
Report 1
Report 2
Report 3
Report 4
Visualizations
Build visual
X-axis
Year
Month
Quarter
Y-axis
Sum of Payment_Value
Legend
Add data fields here
Small multiples
Data
DIM_Customer
Age
Customer_City
Customer_Co...
Customer_Co...
Customer_Pos...
Customer_SK
Customer_Trx_...
Effective_End_...
Effective_Start...
First_Order_D...
Gender
Is_Current
Subscribe_Date
Subscriber_ID
DIM_Date
Date_SK
Day
Full_Date
Month
Month_Name
Quarter
datawarehouse • Last saved: Today at 7:58 PM
Page 3 of 4
80%
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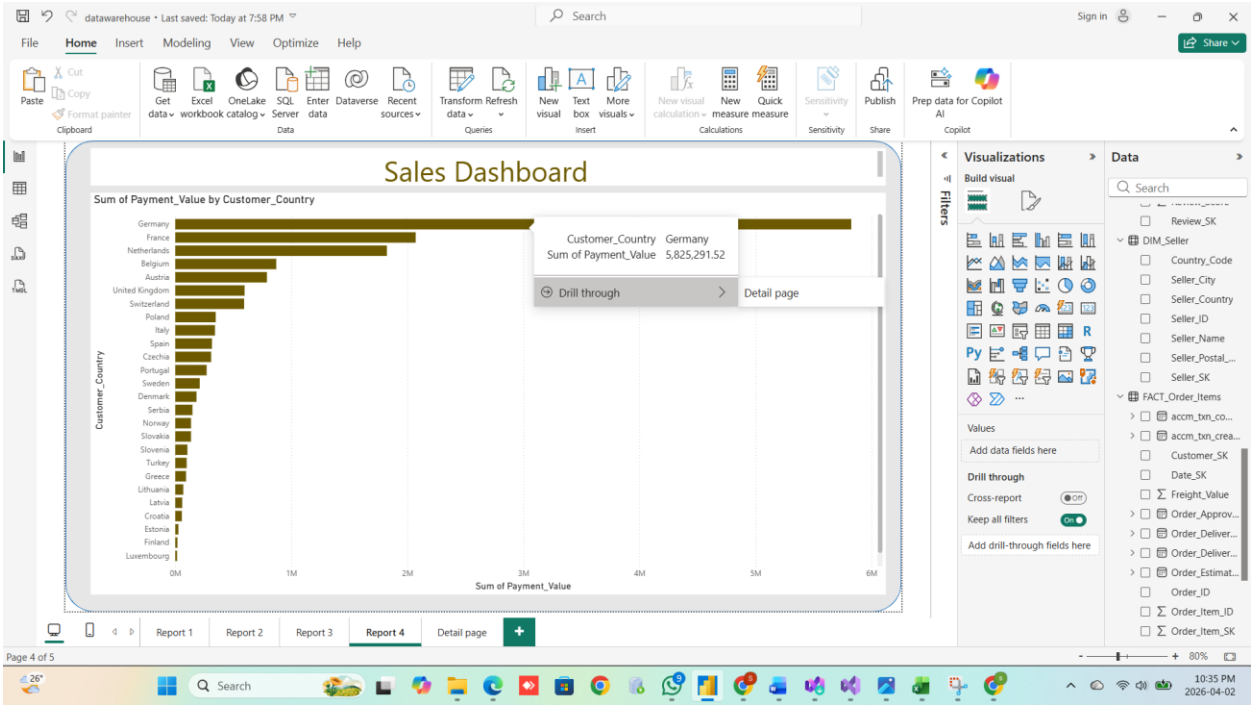
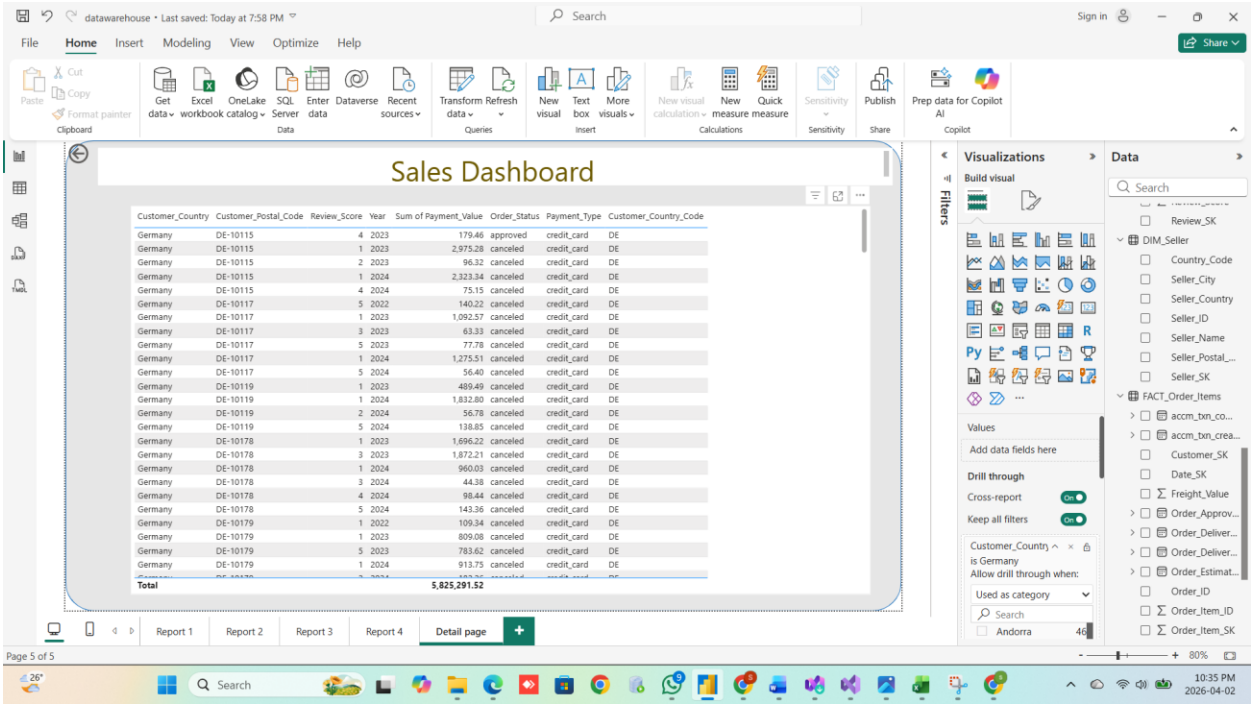
2026-04-02

Report 4

Report 4 implements a drill-through navigation feature across two pages. The first page contains a Horizontal Bar Chart displaying Sum of Payment Value by Customer Country, providing a high-level summary across all European markets. Users can right-click on any country bar, which reveals a tooltip with the country's total payment value along with a Drill Through option to navigate to the Detail page. The Detail page then automatically filters and displays granular transaction-level data including Customer Postal Code, Review Score, Year, Order Status, and Payment Type for the selected country, with a back navigation button allowing users to return to the overview seamlessly.




datawarehouse • Last saved: Today at 7:58 PM
Sales Dashboard
Customer_Country Customer_Postal_Code Review_Score Year Sum of Payment_Value Order_Status Payment_Type Customer_Country_Code
Germany DE-10115 4 2023 179.46 approved credit_card DE
Germany DE-10115 1 2023 2,975.28 canceled credit_card DE
Germany DE-10115 2 2023 96.32 canceled credit_card DE
Germany DE-10115 1 2024 2,323.34 canceled credit_card DE
Germany DE-10115 4 2024 75.15 canceled credit_card DE
Germany DE-10117 5 2022 140.22 canceled credit_card DE
Germany DE-10117 1 2023 1,092.57 canceled credit_card DE
Germany DE-10117 3 2023 63.33 canceled credit_card DE
Germany DE-10117 5 2023 77.78 canceled credit_card DE
Germany DE-10117 1 2024 1,275.51 canceled credit_card DE
Germany DE-10117 5 2024 56.40 canceled credit_card DE
Germany DE-10119 1 2023 489.49 canceled credit_card DE
Germany DE-10119 1 2024 1,832.80 canceled credit_card DE
Germany DE-10119 2 2024 56.78 canceled credit_card DE
Germany DE-10119 5 2024 138.85 canceled credit_card DE
Germany DE-10178 1 2023 1,696.22 canceled credit_card DE
Germany DE-10178 3 2023 1,872.21 canceled credit_card DE
Germany DE-10178 1 2024 960.03 canceled credit_card DE
Germany DE-10178 3 2024 44.38 canceled credit_card DE
Germany DE-10178 4 2024 98.44 canceled credit_card DE
Germany DE-10178 5 2024 143.36 canceled credit_card DE
Germany DE-10179 1 2022 109.34 canceled credit_card DE
Germany DE-10179 1 2023 809.08 canceled credit_card DE
Germany DE-10179 5 2023 783.62 canceled credit_card DE
Germany DE-10179 1 2024 913.75 canceled credit_card DE
Total 5,825,291.52
Report 1
Report 2
Report 3
Report 4
Detail page
Page 5 of 5
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Customer_Country is Germany
Allow drill through when:
Used as category
Andorra
Data
Review_SK
DIM_Seller
Country_Code
Seller_City
Seller_Country
Seller_ID
Seller_Name
Seller_Postal_...
Seller_SK
FACT_Order_Items
accm_txn_co...
accm_txn_crea...
Customer_SK
Date_SK
Freight_Value
Order_Approv...
Order_Deliver...
Order_Deliver...
Order_Estimat...
Order_ID
Order_Item_ID
Order_Item_SK
10:35 PM
2026-04-02

END